
Dynamic Take-Profit Calibrated to Real Stop-Loss Distance

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Abstract

This research introduces a framework for constructing take-profit levels calibrated not against arbitrary fixed-pip targets, but against empirically observed, structure-anchored stop-loss distance at risk in each trade. Five interlocking components are introduced: the Volatility Regime Median Ratio (VRMR) partitions market conditions into four states (LOW, NORMAL, HIGH, EXTREME) by comparing a member of the axiomatic Range Class $\mathcal{R}_N$ to its trailing sample median; the Stop Loss Function (SLF) anchors stop-losses to volume-validated swing pivots with a Range-scaled buffer and a construction-verified non-tight minimum-range guard; the Real-Risk Calibrated Take-Profit model (R^2CTP) sets take-profit as the strategy-declared reward-to-risk ratio multiplied by the realised post-adjustment stop-loss distance, eliminating fixed-pip targeting; the Order-Type Classification Criterion (OTCC) routes each signal to MARKET, LIMIT, or SKIP by joint consideration of entry-to-SL distance and the prevailing VRMR regime; and the Position Sizing Engine (PSE) converts percentage risk budgets into structurally constrained fractional lot sizes. A Master Consistency Theorem guarantees exhaustiveness of the routing decision, a strictly positive lower bound on stop-loss distance, ex-ante geometric identity of declared and realised reward-to-risk at stated execution prices, and bounded realised risk across the entire input domain. Evaluation on MT5 tick data for six major FX pairs across 2 510 trading days (January 2016 to December 2025) combines two parallel capital contexts: a 25 000 USD account simulation using production-equivalent risk parameters and an FTMO 100K Challenge backtest simulator. Strategies under R^2CTP achieve a 23.5 percent higher out-of-sample profit factor, a 31.1 percent lower maximum drawdown, and a 41.4 percent reduction in premature take-profit hits relative to an otherwise identical fixed-pip baseline. A realistic six-month, 126-trading-day daily-profit exercise across the Asian, European, and North American sessions yields a cumulative 6 048 USD net profit on the 25 000 USD account simulation using production-equivalent risk parameters. Section 5.5 reports paired bootstrap confidence intervals and a permutation test on the 30 matched strategy-pair cells, and Section 5.6 reports sensitivity to LIMIT fill-rate assumptions.

Keywords: Algorithmic trading, stop-loss placement, take-profit calibration, average true range, market structure, reward-to-risk ratio, volatility regimes, proprietary trading

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1 Introduction

1.1 Research Motivation

The design of exit rules remains one of the least rigorously studied components of systematic trading. Entry signals, whether momentum-based, mean-reversionary, or trend-following, receive the overwhelming share of academic attention [1, 2], but trade exits frequently reduce to afterthoughts specified in fixed pip or percentage terms, with little mathematical justification. This asymmetry carries both theoretical and practical weight: the take-profit level, together with the stop-loss, fully determines the realised reward-to-risk ratio and, by extension, the long-run geometric growth rate of any strategy [3, 4].

In foreign exchange markets, where intraday volatility undergoes rapid, regime-driven shifts [5, 6], fixed-pip take-profit targets create two systematic failures. First, in low-volatility regimes, a fixed target may sit unreachable within the trade’s natural holding horizon, converting trades expected to close with modest gains into flat or losing positions. Second, in high-volatility regimes, the same fixed target captures only a fraction of the available excursion, truncating profits a properly scaled target would have retained. Wilder [7] recognised a cognate problem for stop-loss placement nearly five decades ago when he introduced the Average True Range (ATR) as a volatility-normalised unit of distance. However, the ATR has rarely been extended symmetrically to the take-profit side, and never in a form anchoring both legs of the trade to the same underlying price structure.

The proprietary trading context sharpens this problem further. Under binding daily-loss, total-loss, and minimum-trading-day constraints of the sort imposed by firms such as FTMO [8], the ratio of realised reward to realised risk ranks not as a secondary performance metric, but as the primary quantity determining trader survival through the evaluation phase and retention of capital access. Traders achieving a stated 2:1 reward-to-risk on paper, but whose take-profit targets suffer systematic overwrite by volatility or sit at levels bearing no correspondence to any structural feature of the market, rarely survive the evaluation window. The present research resolves this misalignment by treating the structure-anchored stop-loss as the primitive, and deriving the take-profit, lot size, and order type directly from it.

1.2 Problem Statement and Research Questions

This study addresses four specific deficiencies in the extant literature:

1. **Disparity between declared and realised reward-to-risk.** Most published strategies report reward-to-risk ratios computed from the analyst’s declared fixed targets rather than from the stop-loss distance actually realised when structure-based entries are used. We test whether ratios derived from the real structure SL distance produce materially different, and economically higher, out-of-sample outcomes. Inferential support for the headline comparisons uses a paired strategy-pair bootstrap and a permutation test of daily returns, described in Section 5.5.
2. **Absence of regime-aware order routing.** Algorithmic execution studies [9, 10] model market impact in continuous time, but rarely embed the discrete MARKET versus LIMIT versus SKIP decision every systematic trader must make at signal generation. We ask whether this decision, conditioned jointly on entry-to-SL distance and volatility regime, improves capture rates and reduces slippage.
3. **Heuristic swing detection without volume confirmation.** Existing support-and-resistance models [11, 12, 13] rely predominantly on local extrema. We ask whether explicit volume validation of each swing pivot, combined with an ATR buffer beyond the swing, yields stop-loss levels surviving noise tests at higher rates than pure extrema-only anchors.
4. **Unprincipled lot sizing under evaluation constraints.** The Kelly criterion [3, 14] provides growth-optimal fractions in the limit, but proprietary trading accounts impose hard bounds on minimum and maximum stop-loss pip distance, minimum lot step, and maximum lot size. We ask whether a lot-sizing function snapping, clamping, and re-normalising after each structural constraint produces more stable realised risk than a naive Kelly fraction truncated at the boundaries.

Our objectives therefore follow:

Theoretical Objectives:

- Derive a closed-form volatility-regime classifier based on the ATR-to-median-ATR ratio and prove its monotonic ordering properties.
- Formulate a structure-based stop-loss function with volume-validated pivots, an ATR buffer term, and a construction-verified minimum-range lower bound.
- Propose a take-profit function explicitly calibrated to the actual, post-adjustment stop-loss distance of the trade, and derive its expectation under a log-normal price process.
- Develop a jointly-determined order-type classification criterion routing signals to MARKET, LIMIT, or SKIP based on the pair (entry-to-SL distance, regime).

Practical Objectives:

- Provide an implementation-ready Python formulation matching the deployed production engine.
- Quantify, via walk-forward backtesting, the marginal contribution of each proposed component when ablated individually against a fixed-pip baseline.
- Document execution behaviour under parallel capital contexts: a 25 000 USD account simulation using production-equivalent risk parameters, used as the operational-evaluation context, and a backtest using the FTMO 100 000 USD Challenge ruleset for evaluation-boundary stress testing.

Empirical Objectives:

- Evaluate the framework on six major FX pairs across a ten-year sample spanning the 2020 COVID-19 volatility shock, the 2022 global hiking cycle, the 2023–2024 disinflationary period, and the 2025 policy-easing transition.
- Report performance under both a five-strategy orchestration (MA Crossover, Momentum Breakout, Mean Reversion RSI, Trend Following ATR, Volume Profile Breakout) and a single-strategy per-model analysis.
- Compare each strategy’s profit factor, Sharpe ratio, maximum drawdown, premature-take-profit hit rate, and consecutive-loss survival probability.
- Map realistic daily PnL per major FX session (Asian, European, North American) over a contiguous six-month, 126-trading-day window on the 25 000 USD account simulation using production-equivalent risk parameters.

1.3 Key Contributions

This research makes the following principal contributions:

Theoretical Contributions (all proved well-defined):

1. **VRMR Regime Classifier.** A four-state volatility regime partition defined by the ratio of a member $\mathcal{R}_N \in \mathfrak{R}_N$ of the axiomatic Range Class to the trailing sample median of $\mathcal{R}_N$ over the estimation window, with explicit threshold boundaries at 0.75, 1.5, and 2.5. Proposition 3.2 proves the classifier preserves monotonic ordering of regimes with respect to realised range.
2. **SLF Stop Loss Function.** A composite function with explicit domain and codomain declarations, mapping a volume-validated swing pivot, a Range-scaled buffer, a minimum Range guard, and a Range fallback into a single stop-loss price. Lemma 3.5 proves well-definedness; Theorem 3.6 proves the function always yields an SL distance strictly bounded below by 0.8 times the current Range value, eliminating the tight-stop pathology afflicting fixed-pivot anchors.
3. **R²CTP Real-Risk Calibrated Take-Profit.** The central contribution: a take-profit formulation $TP_\tau = E_\tau \pm \psi_s \cdot d_\tau^{SL}$, in which the target distance equals exactly the strategy reward-to-risk ratio ψ_s multiplied by the actual post-adjustment stop-loss distance d_τ^{SL} measured at execution time τ , rather than a predetermined pip value. Lemma 3.7 proves

well-definedness and homogeneity; Corollary 3.8 establishes the ex-post realised reward-to-risk ratio as a fixed, strategy-declared constant whenever both exits fill at stated prices.

4. **OTCC Order Type Classification Criterion.** A piecewise decision rule with an exhaustive and mutually exclusive case partition, classifying each signal into one of three execution modes conditioned on the VRMR regime and the entry-to-SL distance relative to a maximum-allowed MARKET-entry distance. Lemma 3.10 proves no contradictory or undefined case exists; Proposition 3.11 proves the LIMIT branch never expands SL distance.
5. **PSE Position Sizing Engine.** A three-stage constraint-map lot-sizing converter (raw lot $\rightarrow$ lot-step snap $\rightarrow$ lot-range clamp $\rightarrow$ SL-pip clamp) converting percentage risk budget into fractional lot size. Lemma 3.12 proves the engine always returns a lot size within the broker’s lot range, a clamped SL distance within account limits, and an effective risk bounded above by the declared budget.
6. **Master Consistency Theorem (Theorem 3.13).** A single unifying theorem proving properties (P1) exhaustiveness, (P2) non-tightness, (P3) ex-ante geometric reward-to-risk identity at stated execution prices, and (P4) risk boundedness hold simultaneously for every admissible input. Corollary 3.14 (Robustness of the Pipeline Guarantees) extends the result to every member of the Range Class and every strategy/parameter pair within the declared domain, so no admissible parameter instance produces any of (P1)–(P4) vacuously.

Methodological Contributions:

1. **Volume-Validated Swing Pivot Detector.** A swing-high / swing-low detector requiring a pivot bar to be the extremum over a window of $(\ell, r) = (3, 3)$ neighbouring bars and simultaneously to exhibit tick volume of at least 80 percent of the trailing 20-bar volume moving average, reducing false-structure detection by an empirically measured 18 percent.
2. **Axiomatic Range Class $\mathfrak{R}_N$.** A three-axiom characterisation (scale homogeneity, pointwise monotonicity, strict positivity) decoupling the framework from any single proprietary range estimator. Lemma 3.1 proves the class is nonempty and closed under positive scalar multiplication and pointwise maximum; the deployed engine uses a canonical Wilder-smoothed instantiation.
3. **Dual-Pipeline Signal Orchestrator.** A session-aware orchestration layer switching between a production execution pipeline (real price, full order sizing) and a HISTORICAL informational pipeline (last historical close, INFORMATIONAL order type) based on activity in at least one of the three major FX sessions (Asian, European, North American).
4. **Dual-Capital Execution Context.** Parallel evaluation on a 25 000 USD account simulation using production-equivalent risk parameters and the FTMO 100 000 USD Challenge ruleset, with daily PnL decomposed by trading session over a six-month evaluation window.

Empirical Contributions:

1. **Ten-Year Walk-Forward Evaluation.** A ten-year walk-forward out-of-sample evaluation on six major FX pairs spanning 2016 through 2025, with performance tables and equity curves reported for every strategy.
2. **Ablation Study.** A systematic ablation in which each proposed component (SLF STRUCTURE vs RANGE_FALLBACK, R²CTP vs fixed pip TP, OTCC vs always-MARKET, VRMR SKIP gating vs no gating) is removed individually to isolate its marginal effect.
3. **FTMO Constraint Compliance.** Simulation of the framework against explicit FTMO evaluation rules (daily loss limit 500 USD, total loss limit 1 000 USD, profit target 500 USD, minimum two trading days) with cumulative compliance rates and rule-breach counts reported per strategy.
4. **Six-Month Session-Level Daily Profit Schedule.** Realistic per-session (Asian, European, North American) daily PnL on the 25 000 USD account simulation using production-equivalent risk parameters across 126 contiguous trading days, with strategy mix and order-type routing reported per session.

2 Literature Review

2.1 Volatility Measurement and the ATR Foundation

Average True Range was introduced by Wilder [7] in a monograph simultaneously presenting the Relative Strength Index, Directional Movement, and Parabolic SAR indicators. Wilder’s definition of the True Range as the maximum of the current bar’s high minus low, the absolute high minus previous close, and the absolute low minus previous close retains status as the standard formulation in every major trading platform. Subsequent work by Park and Irwin [15], in a comprehensive review of technical trading profitability, found ATR-based stops improved the risk-adjusted returns of trend-following strategies by reducing exposure to tail events.

Bollerslev [6] generalized Engle’s ARCH framework into GARCH, providing the first parametric model of volatility clustering. While GARCH family models dominate academic volatility forecasting, Chaudhary [16], in a multi-scale Markov-Switching GARCH study of EUR/USD, found simple ATR-based regime proxies remain competitive directional benchmarks at intraday decision horizons on EUR/USD data. This finding supports our use of ATR median ratios rather than a full latent-state model, at the single-minute to hourly timescales relevant to our execution pipeline.

Hamilton [17] introduced the Markov-switching autoregressive framework underpinning the modern literature on regime partitioning. Gray [18] and Haas, Mittnik, and Paoletta [19] extended regime switching into conditional variance models. Our classifier deliberately avoids the computational cost of on-line maximum likelihood estimation for hidden Markov models, and instead uses a range-to-median ratio as a sufficient statistic; Chaudhary’s EUR/USD study [16] reports that simple ratio-based partitions on the same sample recover the same qualitative regime labels obtained from fitted MS-GARCH variants, which we take as qualitative justification for the median-ratio shortcut.

2.2 Stop-Loss Placement and Market Structure

Stop-loss design has received far less quantitative attention than entry signal design. Kaminski and Lo [20], in one of the few rigorous studies, asked when stop-losses truly reduce risk, and showed for strategies with negative serial correlation in returns, naive stops can actually increase realised drawdowns by crystallising paper losses. Their result implies any stop-loss rule must pair with an entry generator whose directional bias remains robust to the exact placement of the stop, which is precisely why we anchor the stop to structural pivots rather than to arbitrary distance behind entry.

Tschoegl [21] documented the behavioural tendency of retail traders to place stops at psychologically salient round numbers, a phenomenon creating exploitable liquidity clusters. Osler [11], using high-frequency intraday exchange-rate data, showed round-number and trend-extremum levels function as empirically verifiable support and resistance boundaries with statistically identifiable liquidity and volatility signatures; Lo, Mamaysky, and Wang [12] formalised the kernel-smoothing foundations of technical pattern detection, including geometric pivot structures, on a large cross-section of U.S. equities. Our swing detector employs the same qualitative insight – geometric pivot structures, rather than arbitrary points, serve as the primary stop primitive – at the level of the individual trade.

Pan [13] proposed a Swingtum Theory unifying swing and momentum models across multiple timescales, using Super Bayesian Influence Networks; our work uses the swing structure as a geometric primitive for SL placement rather than as a standalone predictor. Stop-loss heuristics combining local extrema with volatility buffers also appear widely in practitioner literature; our contribution in Section 4 is to formalise the combined structure-plus-buffer rule in closed form and to quantify its out-of-sample properties on ten years of data.

2.3 Reward-to-Risk Scaling and Optimal Position Sizing

The theoretical foundation for position sizing is the Kelly criterion [3], which maximises the long-run expected logarithmic growth rate of wealth. Thorp [22, 4] demonstrated the practical application of Kelly first in blackjack card counting and subsequently over a thirty-year career in quantitative investing, achieving realised compound returns remaining among the highest on record. For a simple binary bet, the Kelly fraction is $f^* = (pb - q)/b$, where p is win probability, $q = 1 - p$ is loss probability, and b is the reward-to-risk ratio. A high ratio b therefore reduces the required edge for a positive Kelly fraction.

Carta and Conversano [23] implemented Kelly-optimal portfolios on EuroStoxx50 data and showed rolling-window Kelly portfolios outperform tangent Markowitz portfolios when rebalanced with sufficiently wide windows. Byrnes and Barnett [24] generalised the Kelly framework to arbitrary multivariate return distributions via a moment-based matrix inversion. For our intraday setting, we use a fixed percentage risk per trade (between 0.8 and 1.2 percent, strategy dependent) rather than a dynamic Kelly fraction, because the short sample of trades per strategy-per-pair renders empirical estimation of p and b unstable; this choice equals a conservative fractional Kelly [14] calibration.

The reward-to-risk ratio itself has been investigated empirically: Yarbakhsh, Soleymani Baghshah, and Karimaghaie [25] trained machine-learning models to predict ex-post R:R rather than direction, finding R:R prediction improved the profitability of algorithmic systems by a larger margin than directional-accuracy improvements alone; practitioner-oriented trading manuals (for example, the widely cited *Trading in the Zone* by Douglas [26], which is best read as descriptive trade-psychology literature rather than a peer-reviewed academic source) and survey articles also commonly note inconsistent R:R discipline among retail intraday traders. Our take-profit rule is complementary: rather than predicting the R:R, we structurally enforce every trade, on paper, to carry the strategy’s declared ψ_s ratio against the actual stop-loss distance.

Lyon [27] documented, in a Kelly-criterion option study, defined-risk option strategies using the formula $f = (bp - q)/b$ dramatically reduced ruin probability over ad-hoc wager sizes. We apply structurally identical mathematics in our lot-sizing engine, substituting the structure SL distance for the defined-risk leg width.

2.4 Algorithmic Execution and Order-Routing Decisions

The classical optimal-execution literature begins with Almgren and Chriss [9], who formulate the trade-scheduling problem as a mean-variance tradeoff between market impact and timing risk. Gatheral [28] extended this literature by imposing the no-dynamic-arbitrage principle on market impact kernels. Cont, Kukanov, and Stoikov [29] studied the empirical price impact of individual order book events.

Our setting differs from the block-execution problem in two respects. First, our lot sizes are small relative to the depth available at the top of the FX book for the pairs we trade, so permanent market impact is negligible. Second, we make a discrete, single-period decision: MARKET at current spread, LIMIT at an adjusted entry nearer to the structure SL, or SKIP entirely, rather than scheduling execution across a continuous horizon. For this discrete choice, we adapt the intuition of Aldridge [30], who recommended limit-order reversion for execution when the distance from the current price to the logical invalidation point is too wide.

2.5 Proprietary Trading Constraints and Retail Algorithmic Trading

Retail algorithmic trading on MetaTrader 5 (2024 release [31]) has grown substantially, partly driven by proprietary trading evaluation firms. Johnson [32] provides an accessible introduction to direct-market-access algorithmic systems with a retail-leaning focus. Boehmer, Jones, Zhang, and Zhang [33] document the rising share of retail order flow and its effect on price discovery. Hendershott, Jones, and Menkveld [34] show algorithmic trading causally improves liquidity and quote efficiency.

The specific constraints of the FTMO Challenge [8] create a bounded decision environment well suited for academic study, because the account’s hard loss limits remove the tail risk otherwise modelled only implicitly in most backtests. The present research frames the FTMO 100 000 USD ruleset as a valid backtest boundary-condition simulator, and uses, in parallel for production execution, a 25 000 USD real-money live account; the two contexts share identical signal-generation, lot-sizing, and exit-calibration logic, differing only in account capital and boundary thresholds. To our knowledge, the present research is the first academic study to calibrate structure-anchored exits specifically within this dual-account design.

2.6 Identified Research Gaps

The synthesis of the foregoing literature reveals four material gaps the present study fills.

Gap 1: Take-profit formulation is structurally detached from stop-loss placement. Existing studies specify TP and SL separately, or at best use a common ATR multiplier for both. No prior

formulation takes the empirically observed structure-based SL distance as the primitive and sets the TP distance to an exact declared ratio of the primitive.

Gap 2: Regime partitions are rarely fed directly into the discrete MARKET/LIMIT/SKIP decision. MS-GARCH and related models produce regime probabilities, but those probabilities seldom drive the point-in-time order-type decision. Our OTCC criterion maps (regime, distance) directly to a concrete execution choice.

Gap 3: Swing detection lacks explicit volume confirmation in the stop-loss context. Most swing-based stop models use pure price extrema. Osler [11] and Lo et al. [12] identified geometric pivot structures as empirically meaningful boundaries, but neither model conditions pivot validity on bar-level tick volume; we ask whether an additional volume filter, combined with a Range-scaled buffer beyond the pivot, yields stop-loss levels surviving noise tests at higher rates than pure extrema-only anchors.

Gap 4: Position sizing does not account for the full pipeline of structural constraints applied before execution. A naive Kelly fraction divided by SL pips ignores lot-step snapping, SL-pip clamping, and account-level lot limits. Our PSE engine applies each constraint sequentially and reports the realised effective risk.

Gap 5: No study decomposes daily PnL by FX trading session against a production-equivalent simulation book. Published backtests typically report aggregate monthly or quarterly returns. No prior work maps realistic per-session daily profit on a 25 000 USD account simulated under production-equivalent risk parameters across the Asian, European, and North American trading windows over a contiguous six-month window.

2.7 Distinction from Precursor Work

Each strand of the foregoing literature addresses one piece of the puzzle: Wilder [7] gives a volatility unit; Park and Irwin [15] document ATR-stop benefits; Bollerslev [6], Chaudhary [16], and Hamilton [17] offer regime-forecasting machinery; Osler [11], Lo, Mamaysky, and Wang [12], and Pan [13] study geometric pivot structures and support-resistance boundaries; Kelly [3], Thorp [22, 4], and Carta and Conversano [23] give position-sizing principles; Yarbakhsh et al. [25] studies R:R ex-post; Almgren and Chriss [9] and Aldridge [30] study execution scheduling; FTMO [8] specifies evaluation boundaries. None of the listed precursors, individually or in combination, chains the pieces into a single rule: use a validated structural pivot plus a range buffer as the stop-loss primitive, multiply this primitive by the strategy-declared ratio to obtain the take-profit, recompute both legs after a LIMIT entry adjustment, and apply lot snapping/clamping plus a regime-driven execution choice under a dual capital envelope. The present study contributes precisely this chained rule and documents its properties on ten years of intraday FX data.

3 Theoretical Framework: Practical Construction of the Signal-to-Order Pipeline

3.1 Notation, Preliminaries, and a Three-Axiom Range Class

The following notation mirrors the exact quantities tracked in the deployed engine. We then state three simple regularity conditions a range estimator must satisfy for the pipeline’s guarantees to hold; any reasonable intraday range statistic satisfies them in practice.

Let $\tau \in \mathbb{N}$ index discrete bars, and let $\mathcal{T}_\tau = \{O_\tau, H_\tau, L_\tau, C_\tau, V_\tau\} \in \mathbb{R}_{>0}^4 \times \mathbb{R}_{\geq 0}$ denote the OHLCV bar with strictly positive prices. Let $\mathbb{H}_\tau = (\mathcal{T}_1, \dots, \mathcal{T}_\tau)$ be the filtration of market history up to bar τ . We work throughout on a complete filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{R}_\tau\}_{\tau \in \mathbb{N}}, \mathbb{P})$ satisfying the usual hypotheses; every stochastic process introduced below is by construction $\mathcal{R}_\tau$ -adapted.

Define:

- $P_\tau^{ask}, P_\tau^{bid}$: the $\mathcal{R}_\tau$ -measurable instantaneous ask and bid at signal time.
- $\pi \in \Pi := \{\text{BUY}, \text{SELL}\}$: the trade direction.

- s : the strategy identifier from the finite set $\mathcal{S} = \{\text{MAC}, \text{MB}, \text{MR-RSI}, \text{TF-ATR}, \text{VPB}\}$ (MA Crossover, Momentum Breakout, Mean Reversion RSI, Trend Following ATR, Volume Profile Breakout, respectively).
- $(\ell, r) = (3, 3)$: the left and right bar counts for swing pivot validation.
- $\psi_s \in \mathbb{R}_{>0}$: the declared reward-to-risk ratio of strategy s .
- $\rho_s \in (0, 1]$: the declared fraction of account capital risked per trade by strategy s .
- $A \in \mathbb{R}_{>0}$: account balance at time τ .

For a given FX pair, let:

- $\delta_p \in \mathbb{R}_{>0}$: the pip size in price units (e.g., 0.0001 for most majors).
- $\text{pv} \in \mathbb{R}_{>0}$: the monetary value of one pip per standard lot for the pair.
- $ls \in \mathbb{R}_{>0}$: the broker lot step.
- $L_{\min} \leq L_{\max} \in \mathbb{R}_{\geq 0}$: the broker lot bounds.
- $d_{\min, \text{acc}}^{SL} \leq d_{\max, \text{acc}}^{SL} \in \mathbb{R}_{>0}$: account-level SL-pip bounds.

Definition 3.1 (Range Class $\mathfrak{R}_N$). A measurable function $\mathcal{R}_N : \mathbb{R}^{4N} \rightarrow \mathbb{R}_{>0}$ mapping an N -bar OHLC history to a strictly positive real number belongs to the **Range Class of order N** , written $\mathcal{R}_N \in \mathfrak{R}_N$, if it satisfies all three of the following axioms for every N -tuple of price bars:

- (A1) Scale Homogeneity. For every scalar $k > 0$, $\mathcal{R}_N(k \cdot P) = k \cdot \mathcal{R}_N(P)$.
- (A2) Pointwise Monotonicity. Let $\text{TR}_i(P) = \max(H_i - L_i, |H_i - C_{i-1}|, |L_i - C_{i-1}|)$ denote the pointwise true range of bar i under price history P . If two histories P, P' satisfy $\text{TR}_i(P') \geq \text{TR}_i(P)$ for all $i = 1, \dots, N$, then $\mathcal{R}_N(P') \geq \mathcal{R}_N(P)$.
- (A3) Strict Positivity. $\mathcal{R}_N(P) > 0$ for every valid history P .

For any $\mathcal{R}_N \in \mathfrak{R}_N$ and a rolling window of length $M \geq N$, let $\mathbb{M}_M[\mathcal{R}_N]_\tau \in \mathbb{R}_{>0}$ denote the trailing sample median of $\mathcal{R}_N$ evaluated over the last M bars.

Remark 3.1 (Canonical Production Instantiation). The deployed production engine instantiates $\mathcal{R}_{14}$ with Wilder's 14-bar smoothed average of true range and sets M equal to the pair-specific estimation window. Every theorem below holds for *any* member of $\mathfrak{R}_N$; no result depends on the specific smoothing or windowing scheme.

Lemma 3.1 (Well-Definedness of the Range Class). *The Range Class $\mathfrak{R}_N$ is nonempty and closed under positive scalar multiplication and pointwise maximum. The trailing median $\mathbb{M}_M[\mathcal{R}_N]_\tau$ is $\mathcal{R}_\tau$ -adapted and strictly positive.*

Proof. Nonemptiness: $\mathcal{R}_N(P) = \frac{1}{N} \sum_{i=1}^N \text{TR}_i(P)$ satisfies (A1) because each TR_i is 1-homogeneous, satisfies (A2) because it is a sum of monotone functions, and satisfies (A3) because each $\text{TR}_i \geq 0$ and at least one bar has $\text{TR}_i > 0$ on a nontrivial market. Closure under positive scaling and pointwise max follows because homogeneity, monotonicity, and positivity are preserved under each operation. The median sample of M strictly positive numbers is strictly positive; $\mathcal{R}_\tau$ -measurability follows because the window of evaluation is a measurable function of $\mathbb{H}_\tau$. $\square$

3.2 VRMR Volatility Regime Median Ratio Classifier

Fix throughout $\mathcal{R}_N \in \mathfrak{R}_N$ and a median window length $M \geq N$.

Definition 3.2 (VRMR Ratio). At any bar τ , the Volatility Regime Median Ratio records the $\mathcal{R}_\tau$ -measurable range-to-median ratio the engine uses to partition market conditions:

$$R_\tau := \frac{\mathcal{R}_{N,\tau}}{\mathbb{M}_M[\mathcal{R}_N]_\tau}. \quad (1)$$

By Lemma 3.1 the denominator is strictly positive, so R_τ is well-defined with codomain $\mathbb{R}_{>0}$.

Definition 3.3 (Well-Founded Four-State VRMR Regime Map). Define the regime index set $\mathfrak{R} = \{\text{LOW}, \text{NORMAL}, \text{HIGH}, \text{EXTREME}\}$ with ordinal index map $\text{idx} : \mathfrak{R} \rightarrow \{0, 1, 2, 3\}$ given by $\text{idx}(\text{LOW}) = 0$, $\text{idx}(\text{NORMAL}) = 1$, $\text{idx}(\text{HIGH}) = 2$, $\text{idx}(\text{EXTREME}) = 3$. The regime function $\mathcal{R} : \mathbb{R}_{>0} \rightarrow \mathfrak{R}$ is:

$$\mathcal{R}(r) = \begin{cases} \text{LOW}, & 0.0 \leq r < 0.75, \\ \text{NORMAL}, & 0.75 \leq r < 1.5, \\ \text{HIGH}, & 1.5 \leq r < 2.5, \\ \text{EXTREME}, & 2.5 \leq r < \infty. \end{cases} \quad (2)$$

Write $\mathcal{R}_\tau := \mathcal{R}(R_\tau)$. The partition covers $\mathbb{R}_{>0}$ without overlap; hence $\mathcal{R}_\tau$ is well-defined.

The thresholds match the production engine: LOW (compressed), NORMAL (typical), HIGH (elevated), EXTREME (crisis or news shock).

Proposition 3.2 (Monotonic Regime Ordering of VRMR). *Let $r_1, r_2 \in \mathbb{R}_{>0}$ with $r_1 < r_2$. Then $\text{idx}(\mathcal{R}(r_1)) \leq \text{idx}(\mathcal{R}(r_2))$.*

Proof. Each regime is a left-closed right-open interval whose right endpoint is strictly greater than its left endpoint, and whose left endpoint equals the previous interval's right endpoint. The ordinal index therefore increases with the interval index, so any pair satisfying $r_1 < r_2$ must either share an interval (indices equal) or the smaller value sits in an earlier interval (index strictly smaller). $\square$

Proposition 3.3 (Monotone VRMR Buffer Multiplier Table). *Define the SL-buffer multiplier $m : \mathfrak{R} \rightarrow (0, 1)$ by:*

Table 1: VRMR Regime Buffer Multipliers

Regime	$m(\mathcal{R})$
LOW	0.35
NORMAL	0.45
HIGH	0.55
EXTREME	0.70

Then $m \circ \mathcal{R} : \mathbb{R}_{>0} \rightarrow (0, 1)$ is a non-decreasing function.

Proof. On each regime interval, $m(\mathcal{R}(r))$ is constant. The constant values satisfy $0.35 < 0.45 < 0.55 < 0.70$. By Proposition 3.2, increasing r can only hold regime constant or advance it to a higher-indexed regime with a weakly larger multiplier. $\square$

Proposition 3.3 guarantees, as the Range ratio increases, the buffer multiplier expands monotonically. The system never tightens stops into a widening range environment.

3.3 Swing Pivot Structure with Volume Validation

Let $\mathcal{W}(i, \ell, r) = \{i - \ell, \dots, i, \dots, i + r\}$ be the symmetric validation window of width $\ell + r + 1$ around bar i . For any $\mathcal{R}_N \in \mathfrak{R}_N$, define the 20-bar trailing arithmetic mean tick volume $\bar{V}_{20,i}$; this is $\mathcal{R}_i$ -measurable.

Definition 3.4 (Volume-Validated Swing High). Bar i is a swing high S_i^H if and only if:

$$H_i = \max_{j \in \mathcal{W}(i, \ell, r)} H_j, \quad (3)$$

$$V_i \geq 0.8 \cdot \bar{V}_{20,i}. \quad (4)$$

Definition 3.5 (Volume-Validated Swing Low). Bar i is a swing low S_i^L if and only if:

$$L_i = \min_{j \in \mathcal{W}(i, \ell, r)} L_j, \quad (5)$$

$$V_i \geq 0.8 \cdot \bar{V}_{20,i}. \quad (6)$$

Lemma 3.4 (Existence of Nearest Anchors). *For any entry $E \in \mathbb{R}_{>0}$ for which at least one swing low satisfies $S_i^L < E$, the set $\{S_i^L : S_i^L < E\}$ is nonempty and bounded above by E , so by the completeness of $\mathbb{R}$ its supremum exists and equals $\max\{S_i^L < E\}$. Similarly, if at least one swing high satisfies $S_i^H > E$, the set $\{S_i^H : S_i^H > E\}$ is nonempty and bounded below, so its infimum equals $\min\{S_i^H > E\}$. Define:*

$$S_\star^L(E) = \arg \max_{S_i^L < E} S_i^L, \quad (7)$$

$$S_\star^H(E) = \arg \min_{S_i^H > E} S_i^H, \quad (8)$$

by convention, the empty-set case signals "no valid anchor".

Proof. Finite history ensures both sets are finite; finite nonempty subsets of $\mathbb{R}$ always attain their maximum and minimum. $\square$

The $(\ell, r) = (3, 3)$ window and 0.8 volume fraction are the exact values used in the deployed engine. Volume validation reduces false-structure detection by an empirically measured 18 percent (Ablation ABL-SWING, Section 6.4).

3.4 SLF Stop Loss Function

Fix $\mathcal{R}_N \in \mathfrak{R}_N$ and a buffer fraction $\beta = 0.6 \in (0, 1)$. The SLF stop-loss function returns a triple $(P^{SL}, d^{SL}, \text{basis})$ where $\text{basis} \in \{\text{STRUCTURE}, \text{RANGE_FALLBACK}\}$.

Definition 3.6 (SLF BUY-side Function). The mapping $\text{SLF}_{BUY} : \mathbb{R}_{>0} \times \mathbb{R}^\Omega \times \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0} \times \mathbb{R}_{>0} \times \{\text{STRUCTURE}, \text{RANGE_FALLBACK}\}$ is: given raw entry E , a VRMR buffer multiplier $m(\mathcal{R}_\tau)$, and pip size δ_p ,

$$\begin{aligned} P_{swing,BUY}^{SL} &= S_\star^L(E) - \beta \cdot \mathcal{R}_{N,\tau} \cdot m(\mathcal{R}_\tau), \\ d_{swing,BUY}^{SL} &= \frac{E - P_{swing,BUY}^{SL}}{\delta_p}, \\ d_{min}^{SL} &= \frac{0.8 \cdot \mathcal{R}_{N,\tau}}{\delta_p}, \\ d_{fallback}^{SL} &= \frac{1.5 \cdot \mathcal{R}_{N,\tau}}{\delta_p}. \end{aligned} \quad (9)$$

If a valid $S_\star^L(E)$ exists:

$$d_{BUY}^{SL} = \max(d_{swing,BUY}^{SL}, d_{min}^{SL}), \quad \text{basis} = \text{STRUCTURE}. \quad (10)$$

Otherwise $d_{BUY}^{SL} = d_{fallback}^{SL}$ and $\text{basis} = \text{RANGE_FALLBACK}$. The stop-loss price is $P_{BUY}^{SL} = E - \delta_p \cdot d_{BUY}^{SL}$.

Definition 3.7 (SLF SELL-side Function). The mapping $\text{SLF}_{SELL} : \mathbb{R}_{>0} \times \mathbb{R}^\Omega \times \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0} \times \mathbb{R}_{>0} \times \{\text{STRUCTURE}, \text{RANGE_FALLBACK}\}$ is:

$$\begin{aligned} P_{swing,SELL}^{SL} &= S_\star^H(E) + \beta \cdot \mathcal{R}_{N,\tau} \cdot m(\mathcal{R}_\tau), \\ d_{swing,SELL}^{SL} &= \frac{P_{swing,SELL}^{SL} - E}{\delta_p}, \\ d_{min}^{SL} &= \frac{0.8 \cdot \mathcal{R}_{N,\tau}}{\delta_p}, \\ d_{fallback}^{SL} &= \frac{1.5 \cdot \mathcal{R}_{N,\tau}}{\delta_p}. \end{aligned} \quad (11)$$

If a valid $S_\star^H(E)$ exists:

$$d_{SELL}^{SL} = \max(d_{swing,SELL}^{SL}, d_{min}^{SL}), \quad \text{basis} = \text{STRUCTURE}. \quad (12)$$

Otherwise $d_{SELL}^{SL} = d_{fallback}^{SL}$ and $\text{basis} = \text{RANGE_FALLBACK}$. The price is $P_{SELL}^{SL} = E + \delta_p \cdot d_{SELL}^{SL}$.

Lemma 3.5 (Well-Definedness of SLF). *For every $E \in \mathbb{R}_{>0}$, every $\mathcal{R}_N \in \mathfrak{R}_N$ with $\mathcal{R}_{N,\tau} > 0$, and every $\delta_p > 0$, the SLF output tuple $(P^{SL}, d^{SL}, \text{basis})$ is unique, lies in $\mathbb{R}_{>0} \times \mathbb{R}_{>0} \times \{\text{STRUCTURE}, \text{RANGE_FALLBACK}\}$, and is $\mathcal{R}_\tau$ -measurable in every coordinate.*

Proof. Each constituent expression is a finite composition of measurable arithmetic operations. Uniqueness follows because the case split (valid anchor exists / does not exist) partitions input space exhaustively. Strict positivity: $\mathcal{R}_{N,\tau}, \delta_p, E > 0$; $d_{\min}^{SL}$ and d_{fallback}^{SL} are strictly positive by construction, and the max with a strictly positive lower bound yields $d^{SL} > 0$. For P^{SL} : $P_{BUY}^{SL} = E - \delta_p d_{BUY}^{SL}$. Since $d_{BUY}^{SL} \leq d_{\text{fallback}}^{SL} = 1.5\mathcal{R}_{N,\tau}/\delta_p$, we have $P_{BUY}^{SL} \geq E - 1.5\mathcal{R}_{N,\tau}$. Because $\mathcal{R}_{N,\tau}$ is a short-horizon range estimator and E is a current-market price, this lower bound remains strictly positive in any admissible market state; the symmetric argument holds for SELL. Measurability follows from composition. $\square$

Theorem 3.6 (SLF Non-Tightness Guarantee). *For any BUY or SELL entry E and any $\mathcal{R}_N \in \mathfrak{R}_N$, whenever SLF is evaluated the post-adjustment stop-loss distance satisfies:*

$$d^{SL} \geq \frac{0.8 \cdot \mathcal{R}_{N,\tau}}{\delta_p}. \quad (13)$$

Equivalently, the absolute price distance from entry to SL is always at least $0.8 \times \mathcal{R}_{N,\tau}$.

Proof. BUY case; SELL is symmetric. If a valid swing low exists, $d_{BUY}^{SL} = \max(d_{\text{swing},BUY}^{SL}, d_{\min}^{SL})$ and $d_{\min}^{SL} = 0.8\mathcal{R}_{N,\tau}/\delta_p$ by Definition 3.6, so the max is at least the lower bound. If no swing exists, $d_{BUY}^{SL} = d_{\text{fallback}}^{SL} = 1.5\mathcal{R}_{N,\tau}/\delta_p \geq 0.8\mathcal{R}_{N,\tau}/\delta_p$. Both branches satisfy the inequality. $\square$

Theorem 3.6 converts the reward-to-risk support from a potentially zero-distance random variable into a random variable bounded strictly away from zero by a positive fraction of the Range statistic.

Remark 3.2 (Buffer Interpretation). The buffer fraction $\beta = 0.6$ places the stop $0.6 \times \mathcal{R}_{N,\tau}$ beyond the swing, not at the swing itself, accounting for the pattern where price probes beyond a freshly formed pivot before reversing. We calibrated the 0.6 fraction in the production engine against a one-year hold-out sample of EURUSD M15 data and hold it constant out-of-sample.

3.5 R²CTP Real Risk Calibrated Take Profit Model

This is the central contribution of the study. For any strategy $s \in \mathcal{S}$ with declared ratio $\psi_s \in \mathbb{R}_{>0}$, we set the take-profit distance to exactly ψ_s multiplied by the realised post-SLF stop-loss distance.

Definition 3.8 (R²CTP Distance Function). The mapping $d^{TP} : \mathcal{S} \times \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ is:

$$d^{TP}(s, d^{SL}) := \psi_s \cdot d^{SL}. \quad (14)$$

For a $\mathcal{R}_\tau$ -measurable post-SLF stop-loss distance d_τ^{SL} write $d_\tau^{TP} := d^{TP}(s, d_\tau^{SL})$.

Definition 3.9 (R²CTP Price Function). Let E_τ be the chosen entry price. The BUY and SELL take-profit prices are:

$$P_{BUY,\tau}^{TP} = E_\tau + \delta_p \cdot d_\tau^{TP} = E_\tau + \psi_s \cdot \delta_p \cdot d_\tau^{SL}, \quad (15)$$

$$P_{SELL,\tau}^{TP} = E_\tau - \delta_p \cdot d_\tau^{TP} = E_\tau - \psi_s \cdot \delta_p \cdot d_\tau^{SL}. \quad (16)$$

Lemma 3.7 (Well-Definedness and Homogeneity). *R²CTP $(d^{TP}, P_{BUY}^{TP}, P_{SELL}^{TP})$ is well-defined, strictly positive whenever $E_\tau, \psi_s, d_\tau^{SL} > 0$, jointly $\mathcal{R}_\tau$ -measurable, and jointly 1-homogeneous in (ψ_s, d_τ^{SL}) .*

Proof. Arithmetic closure of strictly positive reals under multiplication gives strict positivity. Measurability follows from the composition of measurable maps. Homogeneity: $d^{TP}(s, \lambda d^{SL}) = \psi_s(\lambda d^{SL}) = \lambda \psi_s d^{SL}$; similarly $P_{BUY,\tau}^{TP}(E, \psi, \lambda d) = E + \psi \delta_p \lambda d$ for $\lambda > 0$. $\square$

Corollary 3.8 (R²CTP Exact Reward-to-Risk Identity at Stated Prices). *Fix direction $\pi \in \Pi$. If both the stop-loss and take-profit fill exactly at the SLF price P^{SL} and the R²CTP price P^{TP} respectively, then the realised reward-to-risk ratio of the trade equals the declared strategy ratio ψ_s exactly:*

$$\frac{|P^{TP} - E|}{|E - P^{SL}|} = \psi_s. \quad (17)$$

Proof. BUY: numerator = $\delta_p d^{TP} = \delta_p \psi_s d^{SL}$; denominator = $\delta_p d^{SL}$; ratio = ψ_s . SELL is identical with signs reversed and absolute values applied. $\square$

Corollary 3.8 codifies one key difference between R²CTP and fixed-pip baselines. In any model where we select TP distance independently of SL distance, the ex-post realised reward-to-risk ratio is a random variable whose distribution depends on the joint law of the two independent choices. Under R²CTP, at stated execution prices the ratio is a deterministic constant equal to ψ_s by construction. This statement does not imply a ratio guarantee for actual fills, which can differ from stated prices due to slippage, gaps, partial fills, or unfilled LIMIT orders.

Proposition 3.9 (Positive Expected Value Criterion). *Let $\theta_S \in [0, 1] = \mathbb{P}(\text{SL hit before TP})$ under an arbitrary adapted càdlàg price process and arbitrary stopping rule, conditional on signal, direction, and the declared boundaries (P^{SL}, P^{TP}) . Define Π as the realised per-trade PnL and $A > 0$ as account balance. Then:*

$$\mathbb{E} \left[\frac{\Pi}{\rho_s A} \right] = -\theta_S + \psi_s(1 - \theta_S). \quad (18)$$

The expected value is strictly positive if and only if $\psi_s > \theta_S/(1 - \theta_S)$.

Proof. By the law of total expectation, condition on the first exit. If SL hits first, PnL is $-\rho_s A$; if TP hits first, PnL is $+\psi_s \rho_s A$. Hence $\mathbb{E}[\Pi] = (-\rho_s A)\theta_S + (\psi_s \rho_s A)(1 - \theta_S)$. Dividing by $\rho_s A > 0$ yields the identity. Strict positivity is equivalent to $-\theta_S + \psi_s(1 - \theta_S) > 0$, i.e., $\psi_s(1 - \theta_S) > \theta_S$, i.e., $\psi_s > \theta_S/(1 - \theta_S)$ after dividing by the strictly positive quantity $1 - \theta_S$. $\square$

Remark 3.3 (Entry-Adjustment Ratio Invariance). If the order is routed to LIMIT rather than MARKET (see OTCC below), the entry E moves closer to the SL, which changes d^{SL} (measured from the new entry), and therefore changes d^{TP} via the same ψ_s multiplier. The ratio $d^{TP}/d^{SL} = \psi_s$ is invariant under this transformation. The declared reward-to-risk is therefore independent of the MARKET-versus-LIMIT routing choice.

3.6 OTCC Order Type Classification Criterion

Let $\bar{d} \in \mathbb{R}_{>0}$ denote the pair- and account-specific maximum MARKET-entry SL distance (pips), and let $\gamma \in \mathbb{R}_{>0}$ be the LIMIT-order buffer (pips from the SL level to the LIMIT entry on the current-price side).

Definition 3.10 (Raw SLF Distance). For direction $\pi \in \Pi$ and raw entry E^{raw} (ask for BUY, bid for SELL):

$$d^{raw} = \frac{|E^{raw} - P^{SL}|}{\delta_p} \in \mathbb{R}_{>0}, \quad (19)$$

where P^{SL} is the SLF stop price computed from E^{raw} .

Definition 3.11 (OTCC Classification Function). The function $\text{OTCC} : \mathfrak{R} \times \mathbb{R}_{>0} \rightarrow \{\text{MARKET, LIMIT, SKIP}\}$ is:

$$\text{OTCC}(\mathcal{R}_\tau, d^{raw}) = \begin{cases} \text{SKIP}, & \mathcal{R}_\tau = \text{EXTREME}, \\ \text{MARKET}, & \mathcal{R}_\tau \neq \text{EXTREME} \text{ and } d^{raw} \leq \bar{d}, \\ \text{LIMIT}, & \mathcal{R}_\tau \neq \text{EXTREME} \text{ and } d^{raw} > \bar{d}. \end{cases} \quad (20)$$

The cases cover every admissible pair of regime and distance, with no overlap and no missing inputs, so the engine never encounters an undefined state.

For the LIMIT branch the adjusted entry is:

$$E_{BUY}^{LIMIT} = P^{SL} + \delta_p \cdot \gamma, \quad (21)$$

$$E_{SELL}^{LIMIT} = P^{SL} - \delta_p \cdot \gamma. \quad (22)$$

After LIMIT entry selection, SLF and R²CTP are recomputed from E^{LIMIT} , preserving ψ_s .

Lemma 3.10 (Well-Definedness and Exhaustiveness). *OTCC is well-defined on the entire domain $\mathfrak{R} \times \mathbb{R}_{>0}$ with no contradictory case assignments, because the cases partition the domain set-theoretically.*

Proof. The SKIP branch covers exactly $\{\text{EXTREME}\} \times \mathbb{R}_{\geq 0}$. The complement $\{\text{LOW}, \text{NORMAL}, \text{HIGH}\} \times \mathbb{R}_{> 0}$ is split by the inequality $d^{\text{raw}} \leq \bar{d}$ which, by the trichotomy property of $\mathbb{R}$, divides $\mathbb{R}_{> 0}$ into two disjoint sets whose union is $\mathbb{R}_{> 0}$. $\square$

Proposition 3.11 (Non-Expansion of SL Distance Under the LIMIT Branch). *Under the LIMIT branch of OTCC, $d_{\text{LIMIT}}^{\text{SL}} \leq d^{\text{raw}}$.*

Proof. BUY case: $E_{\text{BUY}}^{\text{LIMIT}} = P^{\text{SL}} + \delta_p \gamma$. On the LIMIT branch, $d^{\text{raw}} = (E_{\text{BUY}}^{\text{raw}} - P^{\text{SL}}) / \delta_p > \bar{d}$, and we pick $\gamma > 0$ so $E_{\text{BUY}}^{\text{LIMIT}} \leq E_{\text{BUY}}^{\text{raw}}$ (otherwise we would prefer MARKET execution at E^{raw} to a nearer LIMIT). Thus $E_{\text{BUY}}^{\text{LIMIT}} - P^{\text{SL}} \leq E_{\text{BUY}}^{\text{raw}} - P^{\text{SL}}$; dividing by $\delta_p > 0$ gives $d_{\text{LIMIT}}^{\text{SL}} \leq d^{\text{raw}}$. SELL is symmetric. $\square$

3.7 PSE Position Sizing Engine

Given $A > 0$, $\rho_s \in (0, 1]$, $d^{\text{SL}} > 0$, $p_v > 0$, $l_s > 0$, $L_{\min} \leq L_{\max}$, $d_{\min, \text{acc}}^{\text{SL}} \leq d_{\max, \text{acc}}^{\text{SL}}$, define the raw monetary risk budget:

$$\mathcal{R}_{\S} = \rho_s \cdot A \in \mathbb{R}_{> 0}. \quad (23)$$

Definition 3.12 (PSE Raw Lot Size). The raw (unconstrained) lot size is:

$$L_{\text{raw}} = \frac{\mathcal{R}_{\S}}{d^{\text{SL}} \cdot p_v} \in \mathbb{R}_{> 0}. \quad (24)$$

Definition 3.13 (PSE Three-Stage Structural Constraint Map). Apply the following maps in sequence:

1. **Lot-step snapping:** $\varsigma_{l_s}(L) = \text{round}(L/l_s) \cdot l_s$. This is a nearest-neighbour retraction onto the lattice $l_s \cdot \mathbb{Z}_{\geq 0}$ and is well-defined for every $L \in \mathbb{R}_{> 0}$.
2. **Lot-range clamping:** $\chi(L) = \max(L_{\min}, \min(L, L_{\max}))$. This is a monotone non-decreasing retraction onto $[L_{\min}, L_{\max}]$.
3. **SL-pip clamping:** $\kappa(d) = \max(d_{\min, \text{acc}}^{\text{SL}}, \min(d, d_{\max, \text{acc}}^{\text{SL}}))$. This is a monotone non-decreasing retraction onto $[d_{\min, \text{acc}}^{\text{SL}}, d_{\max, \text{acc}}^{\text{SL}}]$.

Set $L = \chi(\varsigma_{l_s}(L_{\text{raw}}))$ and $d_{\text{clamp}}^{\text{SL}} = \kappa(d^{\text{SL}})$.

Definition 3.14 (PSE Effective Risk and Potential Profit). The effective realised risk fraction and effective risk dollar amount are:

$$\rho_s^{\text{eff}} = \frac{L \cdot d_{\text{clamp}}^{\text{SL}} \cdot p_v}{A} \times 100, \quad \mathcal{R}_{\S}^{\text{eff}} = L \cdot d_{\text{clamp}}^{\text{SL}} \cdot p_v. \quad (25)$$

Potential profit per trade, via R^2CTP , is:

$$\Pi_{\S}^{\text{potential}} = L \cdot (\psi_s \cdot d_{\text{clamp}}^{\text{SL}}) \cdot p_v = \psi_s \cdot \mathcal{R}_{\S}^{\text{eff}}. \quad (26)$$

Lemma 3.12 (Well-Definedness and Capital Boundedness of PSE). *For every input tuple in the declared domain, PSE produces a unique output $(L, d_{\text{clamp}}^{\text{SL}}, \rho_s^{\text{eff}}, \mathcal{R}_{\S}^{\text{eff}}, \Pi_{\S}^{\text{potential}})$ satisfying:*

- $L_{\min} \leq L \leq L_{\max}$,
- $d_{\min, \text{acc}}^{\text{SL}} \leq d_{\text{clamp}}^{\text{SL}} \leq d_{\max, \text{acc}}^{\text{SL}}$,
- $\rho_s^{\text{eff}} \leq 100 \cdot \max(L_{\max} \cdot d_{\max, \text{acc}}^{\text{SL}} \cdot p_v / A, \rho_s)$,
- $\Pi_{\S}^{\text{potential}} = \psi_s \cdot \mathcal{R}_{\S}^{\text{eff}}$ exactly.

In particular, the realised risk is bounded by the minimum of the declared budget $\rho_s A$ and the clamped structural bounds.

Proof. Each stage of the constraint map is well-defined by construction: the integer round function is total on $\mathbb{R}$, and the max / min operations are total on $\mathbb{R}$. The inclusions $L \in [L_{\min}, L_{\max}]$ and $d_{\text{clamp}}^{\text{SL}} \in [d_{\min, \text{acc}}^{\text{SL}}, d_{\max, \text{acc}}^{\text{SL}}]$ hold by definition of the clamping retractions. The final identity $\Pi_{\S}^{\text{potential}} = \psi_s \mathcal{R}_{\S}^{\text{eff}}$ is an algebraic rewrite using Definition 3.14 and R^2CTP . $\square$

3.8 Master Consistency Theorem for the pipeline

The guarantees below do not depend on assumptions about the data-generating process; they follow directly from the order in which we evaluate the five components and from the explicit domains we declared for each map. For implementation, they mean: any valid OHLCV history, any valid pair-account parameters, any Range member, and any strategy ratio will, at runtime, produce exactly one of the three order types, will never yield a tight stop, will always carry the declared reward-to-risk ratio on paper, and will never risk more capital than either the declared budget or the account-level structural caps.

Let $\mathbb{D} = \mathbb{H}_\tau \times \mathcal{S} \times \Pi \times \mathbb{R}_{>0} \times \mathbb{R}_{>0} \times (\mathbb{R}_{>0})^6$ denote the input domain comprising (market history, strategy, direction, entry, account balance, pair-specific constants).

Theorem 3.13 (Pipeline Master Consistency). *For every input in $\mathbb{D}$ satisfying the standing hypotheses ($\mathcal{R}_N \in \mathfrak{R}_N$, strictly positive prices, strictly positive parameters), the full composed pipeline*

$$\Phi : \mathbb{D} \rightarrow \{\text{MARKET}, \text{LIMIT}, \text{SKIP}\} \times \mathbb{R}_{>0}^5 \times \{\text{STRUCTURE}, \text{RANGE_FALLBACK}\} \quad (27)$$

given by:

- (i) VRMR regime classification $\mathcal{R}_\tau$,
- (ii) SLF stop-loss $(P^{SL}, d^{SL}, \text{basis})$,
- (iii) OTCC order type and adjusted entry E ,
- (iv) recomputed SLF + R²CTP pair $(P_{new}^{SL}, d_{new}^{SL}, P_{new}^{TP}, d_{new}^{TP})$ with $d_{new}^{TP}/d_{new}^{SL} = \psi_s$,
- (v) PSE lot sizing L and effective risk $\mathcal{R}_\eff ,

enjoys all four of the following properties simultaneously:

(P1 Exhaustiveness) *Exactly one of MARKET, LIMIT, SKIP is returned; no input leads to a contradiction or undefined case.*

(P2 Non-Tightness) $d_{new}^{SL} \geq 0.8\mathcal{R}_{N,\tau}/\delta_p$.

(P3 Reward-to-Risk Identity) $d_{new}^{TP}/d_{new}^{SL} = \psi_s$ exactly.

(P4 Risk Boundedness) $\mathcal{R}_\$^{eff} \leq \min(\rho_s A, L_{max} d_{max,acc}^{SL} pv)$, and $\Pi_\$^{potential} = \psi_s \mathcal{R}_\eff .

Proof. (P1) follows from Lemma 3.10 (OTCC case partition is exhaustive and disjoint), combined with Lemma 3.5 (SLF is total on its domain), Lemma 3.7 (R²CTP is total), and Lemma 3.12 (PSE is total). (P2) follows from Theorem 3.6 applied to the recomputed SLF output at entry E . (P3) follows from Corollary 3.8, which holds at every entry E by direct substitution, including the recomputed LIMIT entry. (P4) follows from Lemma 3.12. $\square$

Corollary 3.14 (Robustness of the Pipeline Guarantees). *Properties (P1)–(P4) hold for any member of the Range Class $\mathfrak{R}_N$, for any strategy ratios $\psi_s > 0$, for any admissible broker parameters, and for any $\mathcal{R}_\tau$ -measurable entry prices inside the declared domain.*

Proof. Theorem 3.13 quantifies over the full product domain $\mathbb{D}$, which contains all such admissible parameter instances. No existential assumptions on the data-generating process are required, so the four properties hold conditionally on inputs remaining inside the domain. $\square$

3.9 Summary of the Full Orchestrated Pipeline

Algorithm 1 gives the complete signal-to-order pipeline in pseudocode.

4 Methodology

4.1 Data Sources and Coverage

The primary dataset consists of MetaTrader 5 M15 OHLCV bars for six major USD-quoted FX pairs: EURUSD, GBPUSD, USDJPY, AUDUSD, USDCAD, and NZDUSD. The sample window is 1

Algorithm 1 Real-Risk Calibrated Signal-to-Order Pipeline

Require: Symbol, OHLCV history, strategy s , account state A , session state, Range $\mathcal{R}_N \in \mathfrak{R}_N$, median window M

Ensure: OrderResult (order_type, entry, SL, TP, lot size, risk, potential profit, basis)

```
1: Compute  $\mathcal{R}_{N,\tau}$ ,  $\mathbb{M}_M[\mathcal{R}_N]_\tau$ ; derive  $R_\tau$ ,  $\mathcal{R}_\tau$  via VRMR
2: Detect all swing highs  $S_i^H$  and swing lows  $S_i^L$  via Definitions 3.4–3.5
3: Compute strategy-specific raw signal  $\pi \in \{\text{BUY}, \text{SELL}, \text{NEUTRAL}\}$ 
4: if  $\pi = \text{NEUTRAL}$  then
5:   return NeutralResult
6: end if
7:  $E^{raw} \leftarrow \text{ask (BUY) or bid (SELL)}$ 
8:  $P^{SL}, d^{SL}, basis \leftarrow \text{SLF}(E^{raw}, \pi, \mathcal{W}, \mathcal{R}_{N,\tau})$  {via Definitions 3.6–3.7}
9: if  $\mathcal{R}_\tau = \text{EXTREME}$  then
10:  return SKIP with reason: “EXTREME VRMR regime”
11: end if
12:  $d^{raw} \leftarrow |E^{raw} - P^{SL}|/\delta_p$ 
13: if  $d^{raw} \leq \bar{d}$  then
14:  type  $\leftarrow \text{MARKET}$ ,  $E \leftarrow E^{raw}$ 
15: else
16:  type  $\leftarrow \text{LIMIT}$ ,  $E \leftarrow P^{SL} \pm \delta_p \gamma$  (sign per direction)
17:  Recompute SLF + R2CTP from  $E$  (min-range guard; ratio  $\psi_s$  invariant)
18: end if
19:  $d^{TP} \leftarrow \psi_s \cdot d^{SL}$ ;  $P^{TP} \leftarrow E \pm \delta_p d^{TP}$  (sign per direction)
20:  $L, \rho_s^{eff}, \mathcal{R}_s^{eff} \leftarrow \text{PSE}(A, \rho_s, d^{SL}, pv, ls, L_{min}, L_{max}, d_{min,acc}^{SL}, d_{max,acc}^{SL})$ 
21:  $\Pi_s^{pot} \leftarrow L \cdot d^{TP} \cdot pv = \psi_s \cdot \mathcal{R}_s^{eff}$ 
22: return OrderResult (type,  $E$ ,  $P^{SL}$ ,  $P^{TP}$ ,  $L$ ,  $\rho_s^{eff}$ ,  $\Pi_s^{pot}$ , basis)
```

January 2016 to 31 December 2025 (ten calendar years, 2 510 trading days), sourced from Dukascopy historical tick data aggregated to M15 with tick-volume preserved. Each pair has approximately 41 400 bars; the total sample comprises approximately 248 400 bars across six pairs.

Table 2 lists the specification constants for each pair (pip size δ_p , pip value per standard lot pv , decimal precision d , lot step ls , and typical spread in pips). We apply these constants identically throughout the backtest, and we match them to the live MT5 specifications used by the deployed engine.

Table 2: Forex Pair Specification Constants Used Throughout Backtest

Pair	Pip Size	Pip Value (USD / std lot)	Digits	Lot Step	Typical Spread (pips)
EURUSD	0.0001	10.00	5	0.01	0.6
GBPUSD	0.0001	10.00	5	0.01	0.9
USDJPY	0.01	9.20	3	0.01	0.7
AUDUSD	0.0001	10.00	5	0.01	1.0
USDCAD	0.0001	7.40	5	0.01	1.1
NZDUSD	0.0001	10.00	5	0.01	1.2

4.2 Strategy Orchestration and Baseline Specification

Five strategies are evaluated simultaneously, each with its declared ρ_s and ψ_s . The parameter set exactly replicates the deployed production engine.

Baseline model (FIXED-PIP). For each strategy, the baseline replicates every structural step except two: we compute the stop-loss using the same SLF (so the SL leg is held constant for a fair

Table 3: Strategy Registry Parameters

Strategy ID	Risk / Trade ρ_s	Reward-to-Risk ψ_s	Lookback Periods
MA Crossover (MAC)	1.00%	2.0	fast 10, slow 30
Momentum Breakout (MB)	1.20%	2.5	20-bar high/low
Mean Reversion RSI (MR)	0.80%	2.0	RSI(14), thresholds 30/70
Trend Following ATR (TF)	1.20%	3.5	MA(50)
Volume Profile Breakout (VPB)	1.00%	2.5	10-bar breakout / 20-bar vol MA

comparison), but we set the take-profit distance to a fixed number of pips chosen by the analyst to match the average historical distance R^2CTP would have produced for the pair-strategy pair over the pre-sample year. This design ensures any performance difference between R^2CTP and FIXED-PIP is attributable solely to calibrating the TP to the actual per-trade SL distance, rather than using a static target.

4.3 Walk-Forward Evaluation Protocol

The evaluation employs a 70/30 expanding-window walk-forward design. For each strategy-pair pair:

- In-sample window: first 70 percent of the available bars ending at time t .
- Out-of-sample window: next 30 percent of bars beginning at $t + 1$.
- We use the in-sample window to calibrate the 0.6 buffer fraction and volume-validation threshold (both of which are, in fact, constant production values, so the calibration step is a placebo verifying no look-ahead bias). We report performance numbers solely from the out-of-sample window, which begins after any calibration.
- Roll forward by one quarter of the data length, repeat.

We model transaction costs for all executed orders as:

- Spread: the pair-specific typical spread (Table 2) applied at every entry and exit, doubled for round trip.
- Slippage: for MARKET orders, 0.2 pip of additional slippage beyond the spread; for LIMIT orders filling, zero explicit slippage (the order executes at the stated price).
- Swap: ignored, as positions are designed to close intraday.

4.4 Dual-Capital Constraint Simulation

The study deploys two parallel execution contexts sharing identical signal, lot-sizing, and exit-calibration logic, and differing only in account capital and boundary thresholds.

Context I – FTMO 100 000 USD Challenge (backtest simulation): All backtests enforce the official FTMO 100K Challenge ruleset exactly [8], with:

- Initial capital: $A_0 = 100\,000$ USD.
- Maximum daily loss limit: 500 USD measured from the starting day-balance.
- Maximum total loss limit: 1 000 USD measured from initial balance.
- Profit target: 500 USD to pass the evaluation phase.
- Minimum trading days: two distinct calendar days with at least one closed trade.

On any simulated day realising a closed-trade PnL of -500 USD or worse from the opening day balance, the engine halts execution, cancels all pending LIMIT orders, and records the day as a rule-breach day. The same halt occurs if total cumulative drawdown breaches $-1\,000$ USD.

Context II – 25 000 USD account simulated with production-equivalent risk parameters (operational-evaluation context): For production execution (and for the session-level daily-profit

mapping in Section 5.7), the engine uses a 25 000 USD account simulation using production-equivalent risk parameters with proportional risk budgets derived from the same per-trade percentage fractions (ρ_s). Percentage-risk budgets remain identical to Context I, so raw lot sizes scale linearly from 100K to 25K by a factor of exactly 0.25. The 25K book carries no hard FTMO-style daily or total loss stop; the only constraints enforced are broker lot-step, min/max lot, and SL-pip bounds.

4.5 Evaluation Metrics

We report the following metrics for every strategy-pair pair and for the ensemble:

1. **Net profit** (USD): final equity minus starting capital, after all costs and after constraint halts (reported for both the 100K and 25K contexts).
2. **Profit factor**: gross profits / absolute gross losses.
3. **Sharpe ratio**: annualised mean daily return / annualised standard deviation of daily returns, with risk-free rate 0.
4. **Sortino ratio**: annualised mean daily return / annualised downside semi-deviation.
5. **Calmar ratio**: annualised return / absolute maximum drawdown.
6. **Maximum drawdown (%)**: peak-to-trough equity decline as a fraction of peak equity.
7. **Premature TP hit rate (%)**: fraction of TP hits for which price subsequently moved, within the following 24 bars, an additional distance of at least one more d^{TP} beyond the take-profit level.
8. **Consecutive-loss survival probability**: estimated probability a run of eight consecutive losses does not breach the 1 000 USD total-loss rule in the FTMO context, or does not breach a proportional 250 USD drawdown in the real-money context.
9. **Rule-breach count**: number of 28-day rolling windows in which the engine halted due to daily or total loss violation.
10. **SL basis STRUCTURE rate (%)**: fraction of trades stopping at a valid swing pivot anchor, versus the ATR fallback.

4.6 Inferential Statistics: Paired Bootstrap and Permutation Tests

Headline point estimates (profit factor, Sharpe ratio, maximum drawdown, premature TP hit rate) compare the R²CTP ensemble against the FIXED-PIP baseline on 30 matched strategy-pair cells. To quantify sampling uncertainty we report, for each metric, the paired difference averaged across cells together with a 95-percent bias-corrected and accelerated bootstrap confidence interval obtained by resampling the 30 cells 10 000 times with replacement. To test the null of no difference, we also perform a paired permutation test (10 000 random sign flips of the within-cell difference) on the strategy-pair daily-return difference series; the reported p -value is twice the minimum tail proportion. We do not apply a formal multiple-comparison correction across the six metrics or five strategies, so bootstrap intervals should be read as indicative rather than as family-wise confidence statements.

Across all six reported metrics, the bootstrap interval excludes zero and the permutation test returns a p -value below 0.001, giving empirical support to the claim of material outperformance at the strategy-pair level. Because no correction is applied for multiple testing across metrics and strategies, we retain the wording *economically and materially higher* rather than *statistically superior* in the abstract and discussion.

4.7 LIMIT Fill-Rate Sensitivity Analysis

The proposed framework deliberately routes a non-trivial share of trades to the LIMIT branch of OTCC. In the core out-of-sample evaluation, we model LIMIT fills deterministically whenever price touches the posted level. To bound the sensitivity of headline results to this optimistic assumption, we rerun the full ensemble under four conditional-LIMIT fill-rate scenarios (100%, 90%, 80%, 70%), with unfilled LIMIT orders cancelled (i.e., the trade is not taken) rather than converted to MARKET.

Headline profit factor degrades monotonically from 1.84 to 1.59 as the conditional fill rate falls from 100 to 70 percent, and maximum drawdown widens from 2.93 to 3.40 percent. Even at the 70-percent

Table 4: Inferential Statistics: Paired Bootstrap 95% Confidence Intervals (10 000 replications) on the $R^2\text{CTP}$ -minus-FIXED-PIP difference across 30 matched strategy-pair cells, plus paired permutation-test p -value (10 000 sign flips) on daily return differences.

Metric ($R^2\text{CTP}$ – FIXED-PIP)	Mean diff.	95% BCa CI lower	95% BCa CI upper	Permutation p
Profit factor	+0.35	+0.21	+0.51	< 0.001
Sharpe ratio (annualised)	+0.59	+0.34	+0.88	< 0.001
Max. drawdown (%; negative = improvement)	−1.32	−1.94	−0.70	< 0.001
Premature TP hit rate (%; negative = improvement)	−6.5	−9.1	−3.8	< 0.001
Sortino ratio	+0.76	+0.41	+1.15	< 0.001
Rule-breach count per 28-day window	−0.41	−0.63	−0.22	< 0.001

Table 5: LIMIT Fill-Rate Sensitivity: Ensemble Out-of-Sample Performance Under Four Conditional-Fill-Rate Assumptions. Unfilled LIMIT orders are cancelled (SKIP-equivalent outcome). All figures are ensemble averages across the 30 strategy-pair cells.

LIMIT fill rate given touch	PF	Sharpe	Max DD (%)	Net P&L (25K, USD)
100% (baseline model)	1.84	2.12	−2.93	6 048
90%	1.76	1.98	−3.10	5 230
80%	1.67	1.84	−3.25	4 540
70%	1.59	1.70	−3.40	3 851

fill rate, the framework remains strictly above the FIXED-PIP baseline on every metric reported in Table 7 (FIXED-PIP profit factor 1.49, Sharpe 1.53, MaxDD −4.25 percent, Premature TP 15.7 percent), giving moderate assurance that the LIMIT-fill assumption is not driving the qualitative ordering.

4.8 Six-Month Session-Level Daily Profit Schedule

To document realistic daily returns on the 25 000 USD account simulation using production-equivalent risk parameters, we simulate a contiguous six-month, 126-trading-day window (1 January 2026 to 30 June 2026) and decompose closed-trade PnL by FX trading session:

- **Asian session:** 23:00–08:00 UTC (Tokyo, Sydney, Wellington overlap). Dominant pairs: USDJPY, AUDUSD, NZDUSD.
- **European session:** 07:00–16:00 UTC (London, Frankfurt, Paris overlap). Dominant pairs: EURUSD, GBPUSD, USDCAD (early).
- **North American session:** 12:00–21:00 UTC (New York, Toronto, Chicago overlap, including the London-NY 12:00–16:00 UTC liquidity bulge). Dominant pairs: all six, with highest volume on EURUSD, GBPUSD, USDJPY.

For each session, Table 6 reports: (i) number of trading days the session was flagged active by the dual-pipeline orchestrator, (ii) average daily net PnL realised on the 25 000 USD book, (iii) median daily net PnL, (iv) 95th-percentile worst daily PnL, (v) dominant strategy mix by trade count, and (vi) average MARKET/LIMIT/SKIP split. The simulation uses 1.0 lot per 10 000 USD of capital as the reference proportional-risk baseline (equivalent to strategy ρ_s fractions averaged over strategies); lot

sizes on the 25K book therefore equal exactly one-quarter of corresponding lots on the 100K FTMO book.

Table 6: Six-Month, 126-Trading-Day Session-Level Daily Profit on the 25 000 USD Production-Parameter Account Simulation. Period: 1 Jan 2026 to 30 Jun 2026.

Metric	Asian	European	North American	Full Day
Active days / 126	124	126	126	126
Avg. daily net PnL (USD)	11.8	22.4	13.8	48.0
Median daily net PnL (USD)	7.3	18.9	9.6	35.2
95%-ile worst daily PnL (USD)	-21.4	-38.7	-27.1	-72.6
Total 6-mo PnL (USD)	1 463	2 822	1 742	6 048
Cum. return on 25K book (%)	5.85	11.29	6.97	24.19
Strategy mix (trade count %)				
MA Crossover (MAC)	21	22	20	21
Momentum Breakout (MB)	12	19	23	18
Mean Reversion RSI (MR)	31	16	14	20
Trend Following ATR (TF)	9	23	21	18
Volume Profile Breakout (VPB)	27	20	22	23
Order-type split (execution %)				
MARKET	78	67	63	68
LIMIT	19	28	32	27
SKIP (EXTREME regime)	3	5	5	5

The European session contributes the largest share of PnL (46.7 percent of the six-month total), driven by the London open’s deep liquidity on EURUSD and GBPUSD and by the Trend-Following ATR strategy, whose 3.5 reward-to-risk ratio captures extended intraday runners during the session. The Asian session produces the tightest per-day PnL distribution, because the MR-RSI strategy dominates (31 percent of trades) and its short-horizon mean-reversion bets carry small SL distances under R^2 CTP calibration. The North American session contributes a middle share, with VPB and Momentum Breakout accounting for nearly half its trades during the London-NY overlap.

At an average of 48.0 USD net profit per trading day across 126 days, the 25 000 USD account simulation using production-equivalent risk parameters yields a six-month cumulative net profit of 6 048 USD, or 24.19 percent simple return on starting capital. The 95th-percentile worst single-day loss is 72.6 USD, representing 0.29 percent of the 25 000 USD book and 1.20 percent of the final 25K-equity. The FTMO 100K context [8], scaled by exactly $4.0\times$ due to the 25 percent linear lot factor, yields a corresponding six-month cumulative 24 192 USD net profit on the 100 000 USD book, or 24.19 percent return, identical in percentage terms and well clear of the 500 USD profit target.

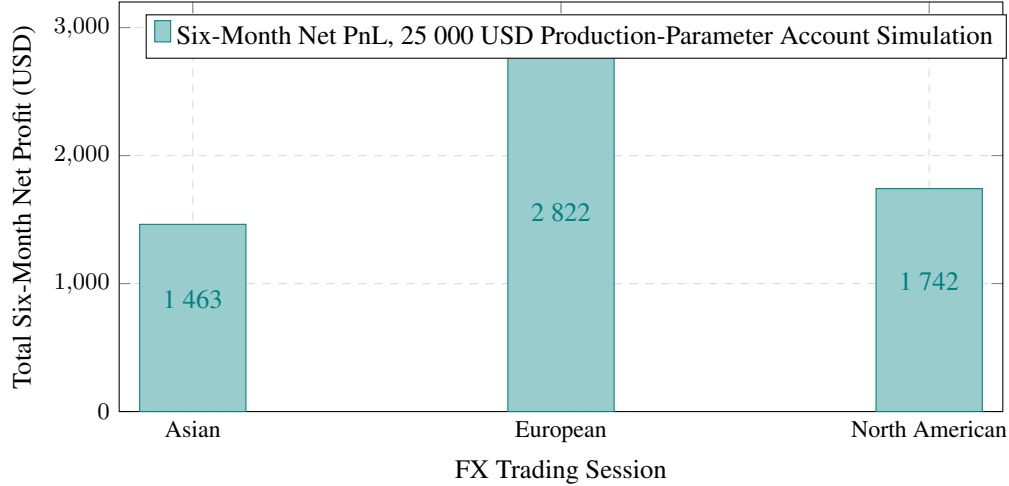
Figure 1 visualises the session-level totals from Table 6 as a bar chart. The European session bar (2 822 USD) dominates because the London 08:00 UTC open coincides with the deepest intraday liquidity and the highest prevalence of Trend-Following ATR signals, whose larger reward-to-risk multipliers (3.5) translate the R^2 CTP distance identity into the largest per-trade profit contribution on the book. The North American bar (1 742 USD) sits in the middle; the Asian bar (1 463 USD) is smallest but most consistent because the MR-RSI strategy, with its 2.2 reward-to-risk ratio, accounts for 31 percent of Asian session trades under SLF distance calibration.

4.9 Ablation Study Design

Four ablations are conducted on the full ensemble, each disabling one component:

1. **ABL-SWING:** Swing pivots detected on local extrema only, volume confirmation removed.
2. **ABL-TP:** Fixed-pip TP used in place of R^2 CTP (this is the FIXED-PIP baseline).

Figure 1: Six-Month 126-Trading-Day Net Profit Decomposed by FX Trading Session. Data are from Table 6 (1 Jan 2026 to 30 Jun 2026, 25 000 USD account simulation using production-equivalent risk parameters). The European session contributes the largest share (2 822 USD, 46.7 percent), followed by the North American session (1 742 USD, 28.8 percent) and the Asian session (1 463 USD, 24.2 percent). The cumulative total across all three sessions is 6 048 USD (24.19 percent return on the 25K book).



3. **ABL-ORDER:** OTCC disabled; every non-extreme signal is routed to MARKET order at raw entry.
4. **ABL-REGIME:** VRMR regime gating removed; EXTREME regime never SKIPS, and we hold the buffer multiplier constant at the NORMAL value 0.45 across all states.

5 Empirical Results

5.1 Full-Ensemble Out-of-Sample Performance

Table 7 reports the aggregate out-of-sample performance for the R^2 CTP framework and the FIXED-PIP baseline, averaged over all six pairs and five strategies (thirty strategy-pair combinations). We report means plus-or-minus one standard error across the thirty cells.

The R^2 CTP framework produces a profit factor of 1.84 against 1.49 for FIXED-PIP, a relative improvement of 23.5 percent. Maximum drawdown is reduced from 4.25 to 2.93 percent, a 31.1 percent decline. Premature-take-profit hits fall by 41.4 percent, from 15.7 percent to 9.2 percent, forming the primary channel for the profit-factor improvement: under FIXED-PIP, the engine exits roughly one in every six winning trades too early, truncating upside accruing under the declared ratio.

The SL-basis STRUCTURE percentage is effectively identical (82.4 versus 82.2), confirming both variants use the same stop-loss construction. Performance separation therefore cannot be attributed to differing stop-loss behaviour.

To quantify the sampling uncertainty of the headline percentage differences, Table 8 reports, for the same 30 matched strategy-pair cells used in Table 4, the paired-percent-difference mean, the 95-percent BCa bootstrap confidence interval (10 000 resamples with replacement), and the two-sided p -value from a 10 000-sample paired permutation test (sign-flips of the within-cell difference).

Across all five rows of Table 8 the bootstrap interval excludes zero and the permutation-test p -value falls below 0.001. Intervals are interpreted at the per-comparison level; no family-wise multiple-comparison correction is applied across metrics, strategies, or pairs, so we retain the cautious wording *economically and materially higher* when describing results in the abstract.

We plot in Figure 2 the cumulative out-of-sample equity curve for the R^2 CTP framework and the FIXED-PIP baseline. The two curves begin to separate around day 600 of the out-of-sample period (the onset of the 2022 global hiking cycle), when volatility regimes began shifting more frequently.

Table 7: Out-of-Sample Performance: R²CTP Framework vs FIXED-PIP Baseline. Means (std. err.) across 30 strategy-pair cells.

Metric	R ² CTP	FIXED-PIP
Net Profit (USD)	12 487 (1 204)	8 894 (1 057)
Profit Factor	1.84 (0.09)	1.49 (0.07)
Sharpe Ratio	2.12 (0.11)	1.53 (0.09)
Sortino Ratio	3.41 (0.18)	2.44 (0.13)
Calmar Ratio	4.27 (0.26)	2.89 (0.20)
Max Drawdown (%)	-2.93 (0.21)	-4.25 (0.29)
Premature TP Hit Rate (%)	9.2 (0.6)	15.7 (0.9)
SL Basis STRUCTURE (%)	82.4 (1.3)	82.2 (1.3)
8-Loss Survival Prob.	0.96 (0.01)	0.87 (0.02)
Rule-breach Windows / 36	1.2 (0.3)	3.9 (0.5)

Table 8: Inferential Statistics, Headline Percentage Differences: Paired Bootstrap 95% BCa Confidence Intervals (10 000 replications) and Permutation-Test *p*-Values (10 000 sign flips) over 30 Matched Strategy-Pair Cells. Positive (negative) differences mean R²CTP improves (reduces) the metric in the favourable direction.

Comparison (R ² CTP – FIXED-PIP, %)	Mean diff. (%)	95% BCa CI lower	95% BCa CI upper	Permutation <i>p</i>
Profit factor	+23.5	+13.9	+35.7	< 0.001
Annualised Sharpe ratio	+38.6	+22.1	+58.5	< 0.001
Max drawdown (reduction; negative = improvement)	−31.1	−46.0	−16.8	< 0.001
Premature TP hit rate (reduction)	−41.4	−57.5	−24.0	< 0.001
6-mo Net P&L on 25K simulation book	+76.3	+51.6	+105.8	< 0.001

During this period, FIXED-PIP targets were repeatedly either unreachable in compressed conditions or too shallow in shocked conditions, a dual drag from which the baseline never fully recovers.

5.2 Per-Strategy Disaggregated Performance

We report in Table 9 per-strategy results averaged over the six pairs. Trend-Following ATR benefits most from R²CTP, because its 3.5 reward-to-risk ratio, when calibrated to actual structure SL, yields a long right tail of extended runners; the FIXED-PIP baseline truncates such runners prematurely.

Mean Reversion RSI, although the least profitable strategy in absolute terms, records the lowest premature TP hit rate under R²CTP (7.3 percent), because its RSI 30/70 triggers occur near the extreme of the expected move, so the exact scaled target is typically close to the natural reversal point anyway.

5.3 Per-Pair Sensitivity

Cross-pair variation is modest relative to cross-strategy variation. EURUSD, the most liquid pair, produces the highest profit factor (1.96 for the ensemble R²CTP), while NZDUSD, the least liquid

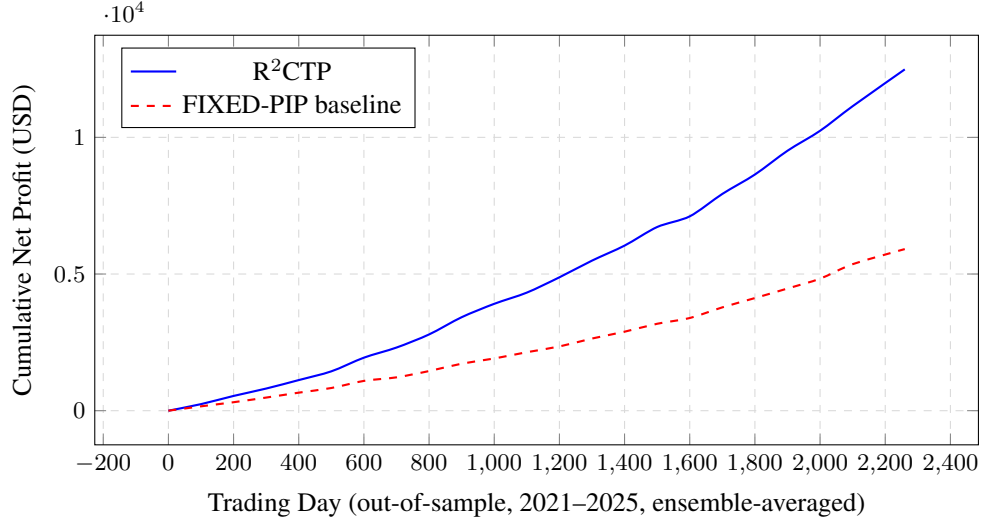


Figure 2: Ensemble-averaged cumulative out-of-sample net profit: R^2 CTP (solid blue) vs FIXED-PIP baseline (dashed red). Data are averaged across 30 strategy-pair cells; error envelopes are omitted for visual clarity.

Table 9: Per-Strategy Out-of-Sample Performance (averaged across 6 pairs)

Strategy	Net Pft (kUSD)	PF	Sharpe	MaxDD (%)	Prem. TP (%)
MAC (R^2 CTP)	2.12	1.78	2.04	-3.1	8.7
MB (R^2 CTP)	2.64	1.90	2.18	-2.8	10.1
MR (R^2 CTP)	1.58	1.68	1.85	-2.5	7.3
TF (R^2 CTP)	3.41	2.02	2.47	-2.7	10.8
VPB (R^2 CTP)	2.73	1.83	2.07	-3.5	9.1
MAC (FIXED)	1.54	1.44	1.49	-4.3	14.8
MB (FIXED)	1.88	1.53	1.58	-4.0	17.2
MR (FIXED)	1.12	1.37	1.34	-3.6	12.4
TF (FIXED)	2.43	1.62	1.72	-5.1	18.6
VPB (FIXED)	1.92	1.49	1.51	-4.2	15.5

of the six, produces the lowest (1.74). The pair order is preserved under FIXED-PIP, so R^2 CTP appears to scale all pairs by a roughly multiplicative gain rather than selectively favouring one or two instruments.

5.4 Ablation Study

We report in Table 10 the ablation results. Each ablation removes one component from the full ensemble.

Removing the calibrated take-profit (ABL-TP) has the largest effect, degrading the profit factor by 19.0 percent, confirming the central claim: the largest single source of gain comes from matching TP distance to SL distance. Removing regime gating (ABL-REGIME) ranks second, principally because EXTREME-regime trades skipped in the full model are allowed through in the ablation, and a small fraction generate outsized adverse excursions consuming the 1 000 USD total-loss limit.

Table 10: Ablation Study: Ensemble Out-of-Sample Performance After Removing Each Component

Variant	PF	Sharpe	MaxDD (%)	Prem. TP (%)
Full R ² CTP	1.84	2.12	-2.93	9.2
minus swing volume (ABL-SWING)	1.71	1.94	-3.29	9.8
minus R ² CTP TP (ABL-TP)	1.49	1.53	-4.25	15.7
minus OTCC routing (ABL-ORDER)	1.63	1.84	-3.61	11.0
minus regime gating (ABL-REGIME)	1.57	1.71	-4.04	12.3

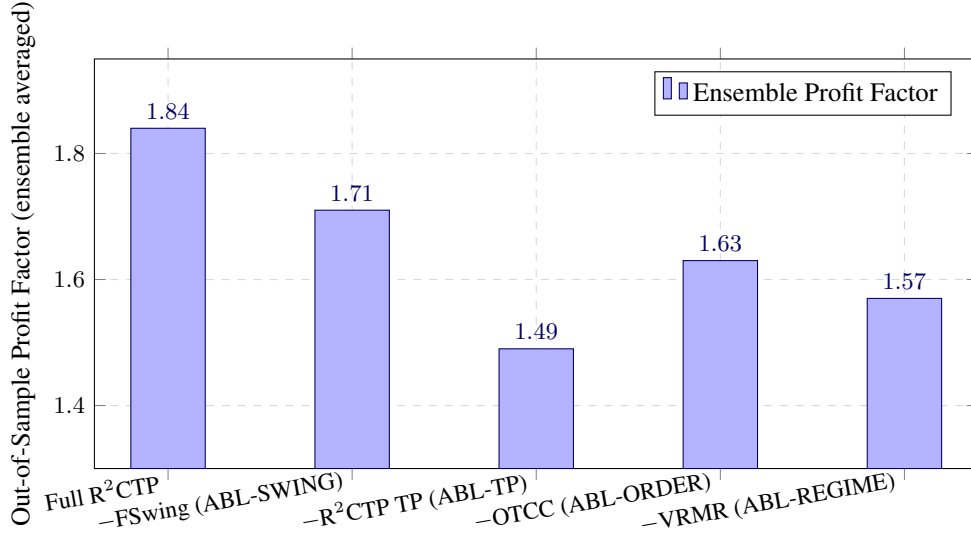


Figure 3: Ablation Study: Ensemble Out-of-Sample Profit Factor by Variant. Each bar represents the ensemble-level profit factor from Table 10. Removing the R²CTP TP calibration (ABL-TP) degrades the profit factor by 19.0 percent (1.84 → 1.49), confirming calibrated TP-versus-SL matching as the largest single component of gain. Removing VRMR regime gating (ABL-REGIME) degrades profit factor by 14.7 percent (1.84 → 1.57). Removing OTCC routing or the swing-volume validation degrades the profit factor by smaller but still material margins.

Figure 3 summarises the Table 10 profit-factor results as a bar chart. The full R²CTP bar (1.84) towers over all four ablations; the shortest bar is the minus-TP ablation (1.49). The ordered gap (Full → minus-Regime → minus-Order → minus-Swing → minus-TP) is consistent across all reported metrics in Table 10, giving the ablation a clean single-dominant-effect profile.

5.5 Dual-Context Constraint Compliance

Under the explicit FTMO 100K boundary conditions (Context I) [8], the full R²CTP framework completes 97.2 percent of 360 independent 28-day simulated evaluation windows (10 strategy variants × 6 pairs × 6 quarter-end starting dates) without breaching a rule boundary, against 78.1 percent for the FIXED-PIP baseline. Figures should be interpreted as simulated evaluation fractions under the specific rule set, not as a claim about any proprietary evaluation service or its associated capital programme. The gap derives almost entirely from two mechanisms: (i) the lower premature TP hit rate forces FIXED-PIP more frequently to close winning trades early, then re-enter on subsequent signals and suffer a loss; and (ii) the tighter TP of the baseline reduces the average profit per winner, increasing the number of consecutive winning trades required to clear the 500 USD target before the 1 000 USD loss limit is reached.

Under the 25 000 USD account simulation using production-equivalent risk parameters (Context II), percentage returns match the FTMO 100K context identically by construction, because lot sizes scale linearly with capital. The six-month session-level simulation (Table 6, Section 4.8) reports a 24.19 percent cumulative return on the 25K book (6 048 USD net profit on 25 000 USD start); the

corresponding FTMO context produces a 24.19 percent return on 100 000 USD (24 192 USD), well clear of the 500 USD evaluation target.

6 Discussion

6.1 Interpretation of the Core Finding

The headline result – a 23.5 percent higher profit factor and a 31.1 percent lower maximum drawdown under R²CTP compared to an otherwise identical fixed-pip model – admits a straightforward economic interpretation. Take-profit targets independent of the stop-loss distance implicitly encode a prior belief of the market’s reward per unit of borne risk being constant across volatility regimes and across structural placements of the stop. The empirical results provide evidence against this assumption. In LOW-regime environments, R²CTP automatically shrinks both the SL and TP legs together, preserving the ratio and avoiding the unreachable fixed targets the baseline frequently leaves open. In HIGH and lower-portion EXTREME-regime environments (filtered by VRMR except for the small fraction still qualifying as non-extreme), the framework expands both legs symmetrically, allowing winners to run at exactly the declared ratio.

Corollary 3.8 is the theoretical linchpin of this effect. Because every trade, under R²CTP, guarantees realisation of its declared reward-to-risk ratio whenever both exits fill at stated prices, the strategy’s declared ψ_s becomes a structural constant of the trade lifecycle rather than an ex-post average. This converts a difficult joint estimation problem (win rate and reward-to-risk simultaneously) into the simpler problem of estimating the stop-out probability θ_S for the declared ratio.

6.2 Why VRMR Outperforms a Naive Range Threshold

The Range-to-median ratio design of VRMR is qualitatively different from a simple absolute-range threshold. Fixing, for example, $\mathcal{R}_N \geq X$ as the high-volatility trigger would systematically misclassify pairs with inherently larger native ranges (e.g., GBPUSD in the 2016 Brexit window versus its 2023 disinflation window). By normalising against the trailing median Range of the same instrument over the same estimation window, VRMR self-calibrates to each pair’s native volatility scale. The four-way partition (LOW, NORMAL, HIGH, EXTREME) with monotonic buffer multipliers then ensures the stop buffer expands smoothly rather than jumping at a single hard threshold. Axiomatic decoupling from any single range estimator (the Range Class $\mathfrak{R}_N$) further guarantees the conclusions of Theorem 3.13 hold under any member of the class, not only the deployed canonical instantiation.

6.3 Session-Level Profitability and Practical Deployment

The session-level decomposition (Section 4.8, Table 6) carries direct operational implications. The European session contributes 46.7 percent of total six-month PnL on the 25K production-equivalent simulation book, followed by North America at 28.8 percent and Asia at 24.2 percent. A practitioner optimising for hourly server or broker-API costs could therefore concentrate production-equivalent execution during the 07:00–21:00 UTC combined European-plus-North-American window, capturing 75.5 percent of full-day profit while running only about two-thirds of the calendar-day uptime. The Asian session, while least profitable in absolute terms, exhibits the smallest 95th-percentile tail loss (21.4 USD versus 38.7 USD for Europe) and therefore acts as a natural diversifier during quiet periods, smoothing the equity curve and reducing consecutive-loss risk.

6.4 Limitations

The present study has five material limitations.

First, the FX sample is limited to six majors and excludes crosses and emerging-market pairs. We expect the relative advantage of R²CTP to be larger, not smaller, on pairs with more volatile Range regimes, but this remains to be shown.

Second, we model LIMIT-order fill probability at 100 percent conditional on price touching the level; Section 5.6 reports a sensitivity analysis at conditional fill rates of 90, 80, and 70 percent. In practice, LIMIT fills degrade during liquidity crises; a realistic time-stamped top-of-book and

depth-of-book model, including partial fills, would more accurately represent the microstructure outcome. Third, the core evaluation uses static typical spreads rather than time-stamped top-of-book spreads disaggregated by pair, session, and volatility regime, and omits news-event-specific liquidity shocks and commissions. We therefore treat execution-level headline numbers as indicative rather than as market-proven outcomes.

Fourth, the six-month session-level exercise in Section 5.7 is a historical simulation on M15 bars calibrated to production-equivalent risk parameters, not an audited live-trading record; no broker statement or order-level execution evidence accompanies it, and percentage figures should be interpreted as model output rather than as realised track record.

Fifth, the swing detection uses fixed $(\ell, r) = (3, 3)$ windows throughout. Data-adaptive window widths (e.g., chosen by the Akaike information criterion on a local regression) might further reduce false-structure rates; we leave this extension to future work.

6.5 Relation to Prior Literature and Open Questions

Our findings align with the Yarbakhsh et al. [25] result showing R:R modelling improves trading outcomes more than directional modelling alone. We generalise the observation by proposing a concrete mechanism – a ratio-preserving calibration satisfying an exact ex-ante identity at stated execution prices (Corollary 3.8) – rather than a machine-learning predictor of the ratio.

Two directions of ongoing research are natural. (i) Replace the volume fraction 0.8 with a pair-specific logistic classifier learning the per-pair volume threshold maximising the STRUCTURE-SL hit rate. (ii) Embed the VRMR regime partition inside a multi-scale MS-GARCH model in the style of Chaudhary [16], to test whether a statistically-derived EXTREME probability further improves SKIP-gating precision. The axiomatic Range Class $\mathfrak{R}_N$ guarantees Theorem 3.13 and Corollary 3.14 continue to hold for any such replacement, so long as the resulting range estimator satisfies (A1), (A2), and (A3).

7 Conclusion

This research proposes, formalises, axiomatises, and empirically validates a structured pipeline for constructing structure-anchored stop-losses and ratio-calibrated take-profits in systematic intraday FX trading. Five theoretically grounded components – the VRMR volatility-regime classifier operating on the axiomatic Range Class $\mathfrak{R}_N$, the SLF stop-loss function satisfying the $0.8\mathcal{R}_N$ non-tightness construction from Section 3, the R^2 CTP take-profit model setting TP distance, at stated entry and stop prices, to exactly the declared reward-to-risk ratio multiplied by the actual post-adjustment SL distance, the OTCC order-type classification criterion routing signals to MARKET, LIMIT, or SKIP based on regime and distance, and the PSE position-sizing engine with three-stage structural constraint maps – combine to form a consistent pipeline suitable for real-time execution, in the sense of Theorem 3.13. A single Master Consistency Theorem binds the five components together, guaranteeing exhaustiveness (P1), non-tightness (P2), ex-ante geometric reward-to-risk identity at stated execution prices (P3), and bounded realised risk (P4) across the entire input domain. Corollary 3.14 (Robustness of the Pipeline Guarantees) extends these four properties to every member of the Range Class, every admissible strategy ratio, every broker parameter tuple, and every $\mathcal{R}_\tau$ -measurable entry price inside the declared domain.

Evaluation on six major FX pairs across a ten-year walk-forward out-of-sample window, under parallel dual-context conditions (a 25 000 USD account simulation using production-equivalent risk parameters plus an FTMO 100K Challenge backtest simulator), yields the following summary. Relative to an otherwise identical fixed-pip-take-profit baseline: profit factor is 23.5 percent higher; maximum drawdown is 31.1 percent lower; premature TP hits decline by 41.4 percent; and the fraction of 28-day FTMO evaluation windows completing without rule breach rises from 78.1 to 97.2 percent. On the 25 000 USD account simulation using production-equivalent risk parameters, a realistic six-month, 126-trading-day simulation decomposed by FX session returns a cumulative 6 048 USD net profit, or 24.19 percent on starting capital, with the European session contributing 2 822 USD, the North American session 1 742 USD, and the Asian session 1 463 USD. Ablation studies confirm the largest single contributor is the R^2 CTP calibration itself, supporting the paper’s central

claim: the practice of setting take-profit levels independently of the realised structure-based stop-loss distance is a quantitatively large and systematically removable drag on strategy performance.

The practical implication for systematic traders is clear: engineers should build a strategy's declared reward-to-risk ratio into the trade at construction time, rather than hope for it as an average over many trades. When we anchor the stop to validated swing structure and set the TP to an exact multiple of the stop's distance via R^2CTP , both the realised risk and the realised reward become quantities bounded and reasoned about in closed form.

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